

PLATFORM COMMUNICATIONS CHANNEL (PCC)

The platform communication channel (PCC) is a generic mechanism for OSPM to communicate with an entity in the platform (e.g. a platform controller, or a Baseboard Management Controller (BMC)). Neither the entity that OSPM communicates with, nor any aspects of the information passed back and forth is defined in this section. That information is defined by the actual interface that employs PCC register address space as the communication channel.

PCC defines a new address space type (PCC Space, 0xA), which is implemented as one or more independent communications channels, or subspaces.

This chapter is arranged as follows:

- The *Platform Communications Channel Table*, *Generic Communications Channel Shared Memory Region*, the *Extended PCC Subspace Shared Memory Region*, and *Reduced PCC Subspace Shared Memory Region* provide reference information about the PCCT, and expected data structures used for the Platform Communications Channel.
- *Doorbell Protocol*, *Platform Notification*, and *Referencing the PCC address space* describe how communications takes place between the OSPM and the platform over PCC.

The PCC interface is described in the following ACPI system description table.

14.1 Platform Communications Channel Table

Table 14.1: Platform Communications Channel Table (PCCT)

Field	Byte Length	Byte Offset	Description
Header			
Signature	4	0	‘PCCT’ Signature for the Platform Communications Channel Table.
Length	4	4	Length, in bytes, of the entire PCCT.
Revision	1	8	2
Checksum	1	9	Entire table must sum to zero.
OEMID	6	10	OEM ID
OEM Table ID	8	16	For the PCCT, the table ID is the manufacturer model ID.
OEM Revision	4	24	OEM revision of PCCT for supplied OEM Table ID.
Creator ID	4	28	Vendor ID of utility that created the table. For tables containing Definition Blocks, this is the ID for the ASL Compiler.
Creator Revision	4	32	Revision of utility that created the table. For tables containing Definition Blocks, this is the revision for the ASL Compiler.
Flags	4	36	Platform Communications Channel Global flags, described in <i>Platform Communications Channel Global Flags</i> .

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Table 14.1 – continued from previous page

<i>Reserved</i>	8	40	<i>Reserved</i>
PCC Subspace Structure[n] (n = subspace ID)	–	48	A list of Platform Communications Channel Subspace structures for this platform. This structure is described in the following section. At most 256 subspaces are supported.

14.1.1 Platform Communications Channel Global Flags

Table 14.2: Platform Communications Channel Global Flags

PCC Global Flags	Bit Length	Bit Off-set	Description
Platform Interrupt	1	0	If set, the platform is capable of generating an interrupt to indicate completion of a command.
<i>Reserved</i>	31	1	Must be zero.

14.1.2 Platform Communications Channel Subspace Structures

PCC Subspaces are described by the PCC Subspace structure in the PCCT table. The subspace ID of a PCC subspace is its index in the array of subspace structures, starting with subspace 0. All subspaces have a common header, followed by a set of type-specific fields:

Table 14.3: Generic PCC Subspace Structure

Field	Byte Length	Byte Offset	Description
Type	1	0	The type of subspace.
Length	1	1	Length of the subspace structure, in bytes. The next subspace structure begins length bytes after the start of this one.
Type specific fields	variable	2	See specific subspace types for more details

This specification defines the following subspaces:

- Type 0, the Generic Communications Subspace,
- Types 1 to 2, HW-Reduced Communications Subspaces,
- Types 3 and 4 are extended PCC subspaces.
- Type 5 is the Hardware Register-Based PCC Subspace.

All other subspace types are reserved.

14.1.3 Generic Communications Subspace Structure (type 0)

Table 14.4: PCC Subspace Structure type 0 (Generic Communications Subspace)

Field	Byte Length	Byte Offset	Description
Type	1	0	0 (Generic Communications Subspace)
Length	1	1	62
<i>Reserved</i>	6	2	Reserved
Base Address	8	8	Base Address of the shared memory range, described in <i>Generic Communications Channel Shared Memory Region</i> .
Memory Length	8	16	Length of the memory range. Must be > 8.
Doorbell Register	12	24	Contains the processor relative address, represented in Generic Address Structure format, of the PCC doorbell. Note: Only System I/O space and System Memory space are valid for values for Address_Space_ID.
Doorbell Preserve	8	36	Contains a mask of bits to preserve when writing the doorbell register.
Doorbell Write	8	44	Contains a mask of bits to set when writing the doorbell register.
Nominal Latency	4	52	Expected latency to process a command, in microseconds.
Maximum Periodic Access Rate	4	56	The maximum number of periodic requests that the subspace channel can support, reported in commands per minute. 0 indicates no limitation.
Minimum Request Turnaround Time	2	60	The minimum amount of time that OSPM must wait after the completion of a command before issuing the next command, in microseconds.

Note

Inaccurate values for the Maximum Periodic Access Rate and Minimum Request Turnaround Time fields can result in punitive side effects for features that rely on the PCC interface. The Platform should report accurate values that allow for maximum channel efficiency while maintaining maximum channel stability.

The Maximum Periodic Access Rate is used by OSPM to determine the maximum rate for periodic evaluation of commands. Infrequent, event driven commands are not restricted by the maximum periodic access rate.

14.1.4 HW-Reduced Communications Subspace Structure (type 1)

The HW-Reduced Communications Subspace is defined in Table 14.5. It is intended for use on HW-Reduced ACPI Platforms, which do not support the SCI. Aside from the interrupt change, and the allowed use of the Functional Fixed HW address space for the Doorbell Register, this subspace is identical to the Generic Communications Subspace described in Section 14.2 and Section 14.5.

Table 14.5: PCC Subspace Structure type 1 (HW-Reduced Communications Subspace)

Field	Byte Length	Byte Offset	Description
Type	1	0	1 (HW-Reduced Communications Subspace)
Length	1	1	62
Platform Interrupt	4	2	GSI of the interrupt used for the PCC platform interrupt for this Subspace.
Platform Interrupt Flags	1	6	Bit [2-7] Reserved Bit [1] Platform interrupt mode 1: Interrupt is Edge triggered 0: Interrupt is Level triggered Bit [0] Platform interrupt polarity 1: Interrupt is Active low 0: Interrupt is Active high
Reserved	1	7	Reserved
Base Address	8	8	Base Address of the shared memory range, described in <i>Generic Communications Channel Shared Memory Region</i> .
Memory Length	8	16	Length of the memory range. Must be > 8.
Doorbell Register	12	24	Contains the processor relative address, represented in Generic Address Structure format, of the PCC doorbell. Note: Only the System I/O, System Memory, and Functional Fixed Hardware spaces are valid for values for Address_Space_ID.
Doorbell Preserve	8	36	Contains a mask of bits to preserve when writing the doorbell register.
Doorbell Write	8	44	Contains a mask of bits to set when writing the doorbell register.
Nominal Latency	4	52	Expected latency to process a command, in microseconds.
Maximum Periodic Access Rate	4	56	The maximum number of periodic requests that the subspace channel can support, reported in commands per minute. 0 indicates no limitation.
Minimum Request Turnaround Time	2	60	The minimum amount of time that OSPM must wait after the completion of a command before issuing the next command, in microseconds.

Note

Inaccurate values for the Maximum Periodic Access Rate and Minimum Request Turnaround Time fields can result in punitive side effects for features that rely on the PCC interface. The Platform should report accurate values that allow for maximum channel efficiency while maintaining maximum channel stability.

The Maximum Periodic Access Rate is used by OSPM to determine the maximum rate for periodic evaluation of commands. Infrequent, event driven commands are not restricted by the maximum periodic access rate.

Type 1 subspaces do not support a level triggered platform interrupt as no method is provided to clear the interrupt. Where level interrupts are required, type 2 or type 3 subspaces should be used.

14.1.5 HW-Reduced Communications Subspace Structure (type 2)

The HW-Reduced Communications Subspace is defined below in Table 14.6. This is intended for use on HW-Reduced ACPI Platforms, which require read-modify-write sequence to acknowledge Platform Interrupt. Aside from three Platform Ack fields at the bottom of the table, this subspace is identical to *HW-Reduced Communications Subspace Structure (type 1)* described above.

Table 14.6: PCC Subspace Structure type 2 (HW-Reduced Communications Subspace)

Field	Byte Length	Byte Offset	Description
Type	1	0	2 (HW-Reduced Communications Subspace)
Length	1	1	90
Platform Interrupt	4	2	GSI of the interrupt used for the PCC platform interrupt for this Subspace.
Platform Interrupt Flags	1	6	Bit [2-7] Reserved Bit [1] Platform interrupt mode 1: Interrupt is Edge triggered 0: Interrupt is Level triggered Bit [0] Platform interrupt polarity 1: Interrupt is Active low 0: Interrupt is Active high
Reserved	1	7	Reserved
Base Address	8	8	Base Address of the shared memory range, described in <i>Generic Communications Channel Shared Memory Region</i> .
Memory Length	8	16	Length of the memory range. Must be > 8.
Doorbell Register	12	24	Contains the processor relative address, represented in Generic Address Structure format, of the PCC doorbell. Note: Only the System I/O, System Memory, and Functional Fixed Hardware spaces are valid for values for Address_Space_ID.
Doorbell Preserve	8	36	Contains a mask of bits to preserve when writing the doorbell register.
Doorbell Write	8	44	Contains a mask of bits to set when writing the doorbell register.
Nominal Latency	4	52	Expected latency to process a command, in microseconds.
Maximum Periodic Access Rate	4	56	The maximum number of periodic requests that the subspace channel can support, reported in commands per minute. 0 indicates no limitation.
Minimum Request Turnaround Time	2	60	The minimum amount of time that OSPM must wait after the completion of a command before issuing the next command, in microseconds.
Platform Interrupt Ack Register	12	62	Contains the processor relative address, represented in Generic Address Structure format, of the platform interrupt ack register. Note: Only the System I/O, System Memory, and Functional Fixed Hardware spaces are valid for values for Address_Space_ID.
Platform Interrupt Ack Preserve	8	74	Contains a mask of bits to preserve when writing the platform interrupt ack register.

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Table 14.6 – continued from previous page

Platform Interrupt Ack Write	8	82	Contains a mask of bits to set when writing the platform interrupt ack register.
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Note

Inaccurate values for the Maximum Periodic Access Rate and Minimum Request Turnaround Time fields can result in punitive side effects for features that rely on the PCC interface. The Platform should report accurate values that allow for maximum channel efficiency while maintaining maximum channel stability.

The Maximum Periodic Access Rate is used by OSPM to determine the maximum rate for periodic evaluation of commands. Infrequent, event driven commands are not restricted by the maximum periodic access rate.

14.1.6 Extended PCC subspaces (types 3 and 4)

Extended PCC communication subspaces are of two types:

Type 3 Initiator subspace: used by the OSPM to communicate with the platform.

Type 4 Responder subspace: Used by the platform to send asynchronous notifications to the OSPM.

Initiator subspaces are not substantially different from type 0, 1, or 2 subspaces. The most notable difference is that a type 3 Initiator subspace does not support asynchronous notifications. Responder subspaces, type 4, provide those notifications, and cannot be used by the OSPM to send messages to the platform.

The format for PCCT entries describing Initiator (type 3), and Responder (type 4) subspaces is shown in the following table.

Table 14.7: PCC Subspace Structure type 3 and type 4

Field	Byte Length	Byte Offset	Description
Type	1	0	3 - Initiator subspace 4 - Responder subspace
Length	1	1	164
Platform Interrupt	4	2	GSI of an interrupt triggered by the platform: For Initiator subspaces (type 3) this is raised when a command is completed on this subspace. For Responder subspaces (type 4) this is raised when platform sends a notification. For a Initiator subspace, this field is ignored if the platform interrupt flag in Table 14.2 is set to zero. If a Responder-subspace is present in the PCCT, then the platform interrupt flag must be set to 1. Note that if interrupts are edge triggered, then each subspace must have its own unique interrupt. If interrupts are level, a GSI may be shared by multiple subspaces, but each one must have unique Platform interrupt Ack preserve and Ack Set masks.
Platform Interrupt Flags	1	6	Bit 7:2 Reserved Bit 1: Platform interrupt mode - Set to 1 if interrupt is Edge triggered - Set to 0 if interrupt Level triggered Bit 0: Platform interrupt polarity - Set to 1 if interrupt is Active low - Set to 0 if interrupt is Active high
Reserved	1	7	Reserved must be zero

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Table 14.7 – continued from previous page

Base Address	8	8	Base Address of the shared memory range, described in Table 14.12 .
Memory Length	4	16	Length of the memory range. Must be ≥ 16 .
Doorbell Register	12	20	Contains the processor relative address, represented in Generic Address Structure (GAS) format, of the PCC doorbell. Note: Only the System I/O, System Memory, and Functional Fixed Hardware spaces are valid values for Address_Space_ID. For Responder subspaces this field is optional, if not present the field should just contain zeros.
Doorbell Preserve	8	32	Contains a mask of bits to preserve when writing the doorbell register.
Doorbell Write	8	40	Contains a mask of bits to set when writing the doorbell register.
Nominal Latency	4	48	Expected latency to process a command, in microseconds. This field is only relevant for Initiator subspaces.
Maximum Periodic Access Rate	4	52	The maximum number of periodic requests that the subspace can support, reported in commands per minute. 0 indicates no limitation. This field is only relevant for Initiator subspaces.
Minimum Request Turnaround Time	4	56	The minimum amount of time that OSPM must wait after the completion of a command before issuing the next command, in microseconds. This field is only relevant for Initiator subspaces.
Platform interrupt Ack Register	12	60	Contains the processor relative address, represented in Generic Address Structure (GAS) format, of the platform interrupt acknowledge register. Note: Only the System I/O, System Memory, and Functional Fixed Hardware spaces are valid for values for Address_Space_ID. If the subspace does not support interrupts or the interrupt is edge driven the register may be omitted. A value of 0x0 on all 12 bytes of the GAS structure indicates the register is not present. If the subspace does support interrupts, and these are level, this register must be supplied. And is used to clear the interrupt by using a read, modify, write sequence.
Platform interrupt Ack Preserve	8	72	Contains a mask of bits to preserve when writing the platform interrupt ack register.
Platform interrupt Ack Set	8	80	Contains a mask of bits to set when writing the platform interrupt ack register.
Reserved	8	88	Reserved must be zero
Command Complete check register address	12	96	Contains the processor relative address, represented in Generic Address Structure (GAS) format, of the Command complete check register. Note: Only the System I/O, System Memory, and Functional Fixed Hardware spaces are valid for values for Address_Space_ID
Command Complete check mask	8	108	Mask to determine whether a command is complete, using the command complete check register. A command is complete if the value of the register when combined through a logical AND with this mask, yields a non-zero value

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Table 14.7 – continued from previous page

Command Complete update register address	12	116	Contains the processor relative address, represented in Generic Address Structure (GAS) format, of the command complete update register. Notes: - Only the System I/O, System Memory, and Functional Fixed Hardware spaces are valid for values for Address_Space_ID. - It is valid for the Command Complete update register to be the same as the Command Complete check register and point to the same address.
Command Complete update preserve mask	8	128	Mask of bits to preserve in the command complete update register, when updating command complete in this subspace.
Command Complete update set mask	8	136	Mask of bits to set in the Command Complete update register, when updating Command Complete in this subspace. For Initiator subspaces, setting this mask results in the Command Complete bit being cleared. For Responder subspaces, setting this mask results in the Command Complete bit being set. Note: OSPM clears those bits in the Command Complete update registers, which are neither set in the Command Complete update preserve mask nor the Command Complete update set mask.
Error status register	12	144	Contains the processor relative address, represented in Generic Address Structure (GAS) format, of the Error status register. This field is ignored by the OSPM on Responder channels Note: Only the System I/O, System Memory, and Functional Fixed Hardware spaces are valid for values for Address_Space_ID Note: this register can be the same as the command complete check register.
Error status mask	8	156	The mask contained here can be combined through a logical AND with content of the Error status register to ascertain whether an error occurred in the transmission of the command through the subspace. The logical NOT of this mask is be used to clear the error. The inverted mask is combined through a logical AND with the content of the Error status register, and the result is written back into said register. This field is ignored for Responder channels.

Note

Inaccurate values for the Maximum Periodic Access Rate and Minimum Request Turnaround Time fields can result in punitive side effects for features that rely on the PCC interface. The Platform should report accurate values that allow for maximum channel efficiency while maintaining maximum channel stability.

Responder subspaces may be used by the platform to send asynchronous notifications to the OSPM. Responder subspace entries in the PCCT share the same format as Initiator subspaces, with the following modifications:

- Type is set to 4 - Responder subspace
- The doorbell may be zero and if so must be ignored by the OSM. If present, the platform can request that the OSPM writes to the doorbell after it has processed a notification.

If a Responder subspace is included in the PCCT, then the global Platform Interrupt flag (see [Table 14.2](#)) must be set to 1.

14.1.7 HW Registers based Communications Subspace Structure (Type 5)

Table 14.8: HW Registers based Communications Subspace Structure (Type 5)

Field	Length (in bytes)	Offset	Value
Type	1	0	5
Length	1	1	Length (includes vendor defined area)
Version	2	2	0x0001 (Version 1 of this PCC definition)
Base Address	8	4	Base Address of the shared memory range, described in Section 14.4
Shared Memory Range Length	8	12	Length of the shared memory range described in Table 14.14 . If this length is zero then based address is ignored.
Doorbell Register	12	20	Contains the processor relative address, represented in Generic Address Structure format, of the PCC doorbell. Note: Only System I/O space and System Memory space are valid for values for Address_Space_ID.
Doorbell Preserve	8	32	Contains a mask of bits to preserve when writing the doorbell register
Doorbell Write	8	40	Contains a mask of bits to set when writing the doorbell register. This is used to send specific commands to the Platform.
Command Complete Check Register	12	48	Contains the processor relative address, represented in Generic Address Structure format, of Command complete Check register. Note: Only System I/O space and System Memory space are valid for values for Address_Space_ID.
Command Complete Check mask	8	60	Contains a mask of bits to query completion status of the previously issued command from the Command Complete Status Register. OS shall do an AND operation with the Command Complete Check Register value. The OS must check the completion status before writing to doorbell register for the next command. If calculated value is 0 then previous operation has been completed. If completion status is not implemented then this mask should be 0. In this case OS shall only wait for Minimum Request Turnaround Time. Note: Command complete check needs be done before writing to doorbell register to avoid any race condition with a previous use of the doorbell register. In addition to that command complete check needs to be done after writing to doorbell register and before reading Status from Error Status Register.
Error Status Register	12	68	Contains the processor relative address, represented in Generic Address Structure format, of Error Status register. Note: Only System I/O space and System Memory space are valid for values for Address_Space_ID.
Error Status mask	8	80	Contains a mask of bits to get error status of the previous command request from Error Status Register. OS shall do an AND operation with Error Status register value. If this mask value is 0 then Error Status register is ignored. Error Status needs to be checked after completion status indicates issued command has been completed. If Command Complete Check is not implemented (means Command Complete Check mask is 0) then wait for Minimum Request Turnaround Time. If the calculated value is zero, then it indicates success. Any other value indicates failure.
Nominal Latency	4	88	Expected latency to process a command, in microseconds.

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Table 14.8 – continued from previous page

Minimum Request Turnaround Time	4	92	The minimum amount of time that OSPM must wait after the completion of a command before issuing the next command, in microseconds.
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14.2 Generic Communications Channel Shared Memory Region

Table 14.9: Generic Communications Channel Shared Memory Region

Field	Byte Length	Byte Offset	Description
Signature	4	0	The PCC signature. The signature of a subspace is computed by a bitwise-or of the value 0x50434300 with the subspace ID. For example, subspace 3 has the signature 0x50434303. The signature is populated by the platform and is verified by OSPM.
Command	2	4	PCC command field, described in the <i>Generic Communications Channel Command Field</i> .
Status	2	6	PCC status field, described in the <i>Generic Communications Channel Status Field</i> .
Communication Space	–	8	Memory region for reading/writing PCC data. The size of this region is 8 bytes smaller than the size of the shared memory region (specified in the General Communications Subspace structure). The first byte of this field represents PCC address 0.

14.2.1 Generic Communications Channel Command Field

For channels of type 0 to 2, this 16-bit field is used to select one of the defined commands for the platform to perform. OSPM is responsible for populating this field before each command invocation.

Table 14.10: Generic Communications Channel Command Field

Field	Bit Length	Bit Offset	Description
Command	8	0	Command code to execute. Command codes are application specific and defined by the consumer of this interface.
<i>Reserved</i>	7	8	Reserved.
Notify on completion	1	15	If set, the platform should generate a Doorbell interrupt at the completion of this command. The interrupt is an SCI for a Type 0 subspace structure, or as described by the Doorbell Interrupt field for Type 1 and Type 2 subspace structures. If the Doorbell bit is not set in the PCC global flags, this bit must be cleared.

14.2.2 Generic Communications Channel Status Field

Table 14.11: Generic Communications Channel Status Field

Field		Bit Length	Bit Off-set	Description
Command Complete	Com- plete	1	0	If set, the platform has completed processing the last command.
Platform interrupt		1	1	If set, the platform has issued a Platform Interrupt to this subspace. OSPM must check the Command Complete and Platform Notification fields to determine the cause of the Interrupt.
Error		1	2	If set, an error occurred executing the last command.
Platform Notification	Notifica- tion	1	3	If set, indicates the platform is issuing an asynchronous notification to OSPM.
Reserved		12	4	Reserved.

Note

OSPM (either in an Interrupt handler or via polling) is required to detect that the Command Complete bit has been set and to clear it before issuing another command. While waiting for this bit to be set, OSPM must not modify any portion of the shared memory region.

Note

The Platform Interrupt bit is required to be cleared in OSPM's Interrupt handler so that a new event can be detected.

14.3 Extended PCC Subspace Shared Memory Region

Table 14.12: Initiator Responder Communications Channel Shared Memory Region

Field	Byte Length	Byte Offset	Description
Signature	4	0	The PCC signature. The signature of a subspace is computed by a bitwise-or of the value 0x50434300 with the subspace ID. For example, subspace 3 has the signature 0x50434303. The signature is populated by the platform and is verified by OSPM.
Flags	4	4	See <i>Initiator Responder Communications Channel Flags</i> below.
Length	4	8	Length of payload being transmitted including command field.
Command	4	12	Command being sent over the subspace.
Communication subspace	–	16	Memory region for reading/writing PCC data. The maximum size of this region is 16 bytes smaller than the size of the shared memory region (specified in the Initiator Responder Communications Subspace structure). When a command is sent to or received from the platform, the size of the data in this space will be Length (expressed above) minus the 4 bytes taken up by the command.

The 32-bit command field is used to select one of the defined commands for the platform to perform. On Initiator subspaces the OSPM is responsible for populating this field, alongside the command's payload, length and flags when sending a command. On command completion, the OSPM is also responsible for interpreting the length and payload fields to ascertain if any response was sent by the platform. For Responder subspaces, the OSPM is responsible for interpreting the command and payload fields to ascertain the nature of the notification that was sent. The format for the flags field is shown in the table below.

Table 14.13: Initiator Responder Communications Channel Flags

Field	Bit Length	Bit Off-set	Description
Notify on completion	1	0	For Initiator subspaces, this field indicates to the platform that it must generate an interrupt when the command has completed. Setting this bit to 1 when sending a command requests that completion of the command is signaled via the platform interrupt. Setting this bit to 0 when sending a command requests that no interrupt is asserted when the command is completed. For Responder subspaces, if the doorbell field of the Responder subspace is non-zero, and this flag is set, the OSPM must access the doorbell once it has processed the notification. For Initiator subspaces, this bit is ignored by the platform if the Platform Interrupt field of the <i>Platform Communications Channel Global Flags</i> is set to zero.
Reserved	31	1	

14.4 Reduced PCC Subspace Shared Memory Region

Table 14.14: Reduced PCC Subspace Shared Memory Region

Field	Byte Length	Byte Offset	Description
Signature	4	0	The PCC signature. The signature of a subspace is computed by a bitwise-or of the value 0x50434300 with the subspace ID. For example, subspace 3 has the signature 0x50434303.
Communication Subspace	–	4	Memory region for reading/writing PCC data. The maximum size of this region is 4 bytes smaller than the size of the shared memory region (specified in the Type 5 PCC Subspace structure). When a command is sent to or received from the platform, the size of the data in this space will be Length (expressed above) minus the 4 bytes taken up by the Signature.

14.5 Doorbell Protocol

Note: This section does not apply to type 4 responder subspaces. Refer to the *Platform Notification* section for information on Type 4 subspace usage.

The doorbell is used by OSPM to notify the platform that the shared memory region contains a valid command that is ready to be processed. A doorbell consists of a hardware register that is accessed via I/O or memory mapped I/O, abstracted in the doorbell field of the PCC subspace structure. OSPM rings the doorbell by performing a read/modify/write cycle on the specified register, preserving and setting the bits specified in the preserve and write mask of the PCC subspace structure.

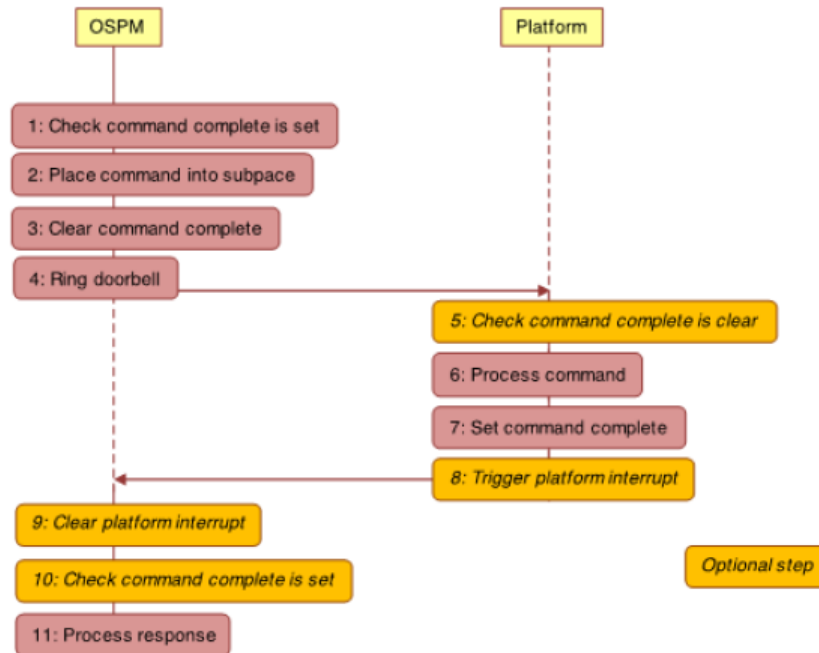


Fig. 14.1: Communication flow of the doorbell protocol

The *Communication flow of the doorbell protocol* (Fig. 14.1) illustrates the steps that the OSPM takes to send a message to the platform over a PCC subspace. It also illustrates the steps the platform takes when it receives the message.

1. First the OSPM checks that there is no command pending completion on the subspace, which is done by checking that the Command Complete bit is set. If Command Complete is set, the subspace is free for use and the shared memory associated with the subspace is exclusively owned by the OSPM. Note: the location of the Command Complete bit differs between subspaces of types 0-2 and those of type 3. For type 0-2 subspaces, the Command Complete bit is in the status register as described in [Section 14.2.2](#). Type 3 subspaces still use a single Command Complete bit, but allow the platform to specify the location and format of the register holding it. Therefore, PCCT structures describing type 3 subspaces use masks and an address to describe how to check the bit. OSPM combines the content of the Command Complete check register, through a bitwise AND of the Command Complete check mask. A non-zero value indicates that the Command Complete bit is set. On type 0 channels, whether the platform sets Command Complete when the subspace is initialized is implementation defined. On these subspaces, the OSPM does not have to check for Command Complete to be set before sending the first command.
2. The OSPM places a command into the shared memory of the subspace, updating the flags, length, command and payload fields (see [Section 14.2](#) and [Table 14.12](#)). If the platform indicates support for platform interrupts in the PCCT (see [Section 14.1.1](#)), then the OSPM can request that the platform generate an interrupt once it has completed processing the command. This is requested by setting the Notify on completion bit in the flags (see

Section 14.1.1 and Table 14.13).

3. The OSPM then clears the Command Complete bit. This step transfers ownership of the shared memory to the platform. Note: clearing the Command Complete bit is done through the Command Complete update register, which can differ in address from the Command Complete check register. In case the Command Complete update register differs in address from the Command Complete check register, the platform must ensure that writes to the Command Complete update register affect the Command Complete check register bits as specified. To reduce platform complexity, it is therefore recommended that the Command Complete update register is the same as the Command Complete check register. To clear the Command Complete bit, the content of the Command Complete update register is combined through a bitwise AND with the Command Complete update preserve mask. The result is then combined through a bitwise inclusive OR with the Command Complete update set mask and the result is written back to the Command Complete update register. As a result, the bits which are neither set in the Command Complete update set mask nor in the Command Complete update preserve mask are implicitly cleared in the Command Complete update register.
4. OSPM rings the doorbell by performing a read/modify/write cycle on the specified doorbell register, preserving and setting the bits specified in the preserve and write mask of the PCC subspace structure.

When the platform receives the command, it executes the following steps:

5. For robustness the platform might optionally check that the Command Complete bit is clear.
6. Processes the command.
7. Sets the command complete bit.
8. Triggers the platform interrupt indicated by the GSI of the subspace's PCCT entry (see Table 14.7). This will only occur if an interrupt has been requested in step 2, and interrupts are supported by the platform. A platform can indicate support for interrupts through the Platform interrupt flag (See Table 14.2)

OSPM can detect command completion either by polling on the Command Complete bit or via platform interrupts. When the OSPM detects that the command has completed, it proceeds with the following steps:

9. If necessary clears platform interrupt. This step applies if:
 - Platform interrupts are supported by the platform on command completion (see Table 14.2).
 - The interrupt was requested by the OSPM through the Notify on completion flag (see Table 14.9 and Table 14.13).
 - The interrupt is described as being a level triggered through the Platform Interrupt flags, and Platform Interrupt Ack register address, and associated masks are provided by the subspace PCCT entry (see table entries for types 2 and 3).
10. If detecting command completion via interrupt, optionally checks that the command is complete. If the platform interrupt is shared among multiple subspaces, this can be used to determine if the interrupt was targeted to this subspace.
11. Processes the command response.

To ensure correct operation, it is necessary to ensure that all memory updates performed by the OSPM in step 2 are observable by the platform before step 3 completes. Equally, all memory updates performed by the platform in step 6 must be observable by the OSPM before step 7 completes.

Note

For subspace types 0 to 2, all accesses to the Status Field must be made using interlocked operations, by both entities sharing the subspace. Types 3-4 avoid this requirement. This requirement will be removed for subspace types 0 to 2 as part of deprecation of platform async notifications in a future spec revision (see Section 14.6).

14.6 Platform Notification

The following sections describe platform notifications on subspace types 0-2 and type 4.

14.6.1 Platform Notification for Subspace Types 0, 1, and 2

The doorbell protocol is a synchronous notification from OSPM to the platform to process a command. If the platform wants to notify OSPM of an event asynchronously, it may set the Platform Interrupt and Platform Notification status bits and issue a Platform Interrupt. OSPM will service the Interrupt, clear the Platform Interrupt and Platform Notification bits, and service the platform notification. The meaning of the platform notification and the steps required to service it are defined by the individual components utilizing the PCC interface.

The platform must wait until OSPM has issued a consumer defined command that serves to notify the platform that OSPM is ready to service Platform Notifications. The command is subspace specific and may not be supported by all subspaces. Platform Notifications must be used in conjunction with an interrupt. Polling for Platform Notifications is not supported.

The platform may not modify any portion of the shared memory region other than the status field when issuing a platform notification.

Platform notifications for subspace types 0, 1, and 2 will be deprecated in a future revision of the specification. Implementers requiring the platform be able to send asynchronous notifications to OSPM should use Initiator/Responder subspaces.

Note: All accesses to the Status Field must be made using interlocked operations, by both entities sharing the subspace. This requirement will be removed for subspace types 0 to 2 as part of deprecation of platform async notifications in a future spec revision.

14.6.2 Platform Notification for Responder PCC subspaces (type 4)

Initiator subspaces (Type 3) only allow synchronous OSPM-initiated communication with the platform, and do not use the platform notification mechanism provided for subspaces of types 0 to 2. Instead, an Initiator subspace can be paired with a Responder subspace, type 4, which is specifically provided for platform-initiated communications with OSPM. Like type 3 Initiator subspaces, type 4 Responder subspaces include a single Command Complete bit. OSPM must set the Command Complete bit when it is ready to receive notifications from the platform.

The flow of communications for a notification is illustrated in [Fig. 14.2](#). As can be seen the communication flow is very similar to that of an Initiator subspace, shown in [Fig. 14.1](#), except that the roles of the platform and the OSPM are reversed.

The steps are as follows:

1. First, the platform checks that there is no notification command pending completion on the subspace. This is done by checking that the Command Complete bit is set. If Command Complete is set the subspace is free for use, and the shared memory associated with the subspace is exclusively owned by the platform.
2. The platform places a notification command into the shared memory in the subspace, updating the flags, length, command, and payload fields (see [Table 14.12](#)). The platform can request that the OSPM rings the doorbell once it has completed processing the notification command by setting the Notify on completion bit in the flags (see [Table 14.13](#)).
3. The platform then clears the command complete bit. This transfers ownership of the shared memory to the OSPM.
4. The platform raises the platform interrupt indicated by the GSI of the Responder subspace.

When the OSPM receives the interrupt, it executes the following steps:

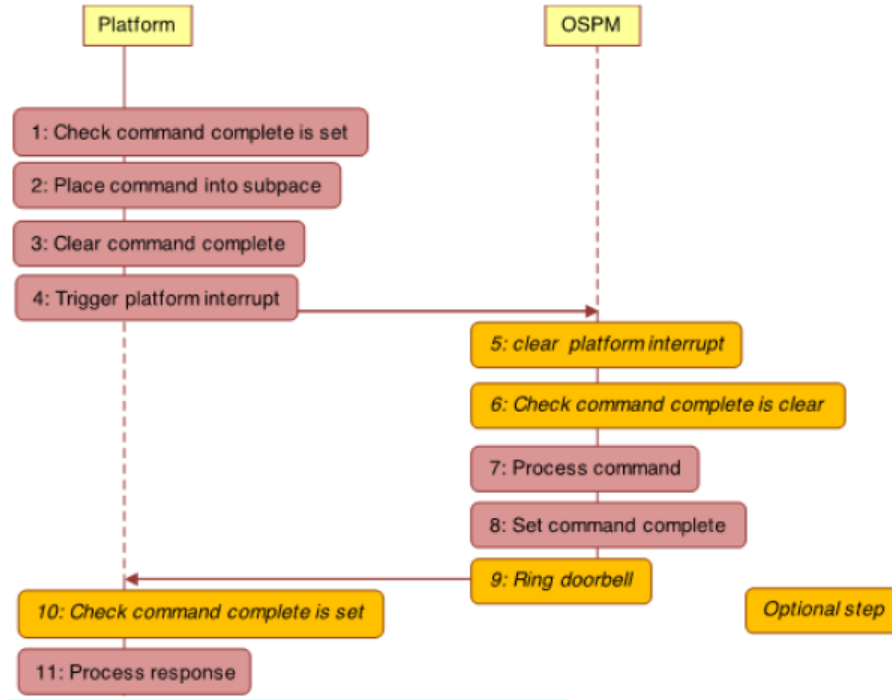


Fig. 14.2: Communication flow for notifications on Responder subspaces

5. Clears the platform interrupt. This is required if the interrupt is described as being a level triggered through the Platform Interrupt flags, and Platform Interrupt Ack register address, and associated masks are provided by the subspace PCCT entry (see Table 14.7).
6. Optionally checks the command complete bit is clear using the Command Complete check register and masks. If the platform interrupt is shared among multiple subspaces, this can be used to determine if the interrupt was targeted to this subspace. Note: Type 4 subspaces use a single Command Complete bit. The PCCT structures describing type 4 subspaces use masks and an address to describe how to check the bit. OSPM combines the content of the Command Complete check register, through a bitwise AND of the Command Complete check mask. A zero value indicates that the Command Complete bit is clear.
7. Processes the notification command.
8. Sets the Command Complete bit using the Command Complete update register and masks. Note: setting the Command Complete bit is done through the Command Complete update register, which can differ in address from the Command Complete check register. In case the Command Complete update register differs in address from the Command Complete check register, the platform must ensure that writes to the Command Complete update register affect the Command Complete check register bits as specified. To reduce platform complexity, it is therefore recommended that the Command Complete update register is the same as the Command Complete check register.

To set the Command Complete bit, the content of the Command Complete update register is combined through a bitwise AND with the Command Complete update preserve mask. The result is then combined through a bitwise inclusive OR with the Command Complete update set mask and the result is written back to the Command Complete update register. As a result, the bits which are neither set in the Command Complete update set mask nor in the Command Complete update preserve mask are implicitly cleared in the Command Complete update register.

9. Rings the doorbell using the doorbell register and masks. This is required if the doorbell ring was requested by the platform in step 2 above. This also requires that the PCCT entry for the subspace has a non-zero doorbell register address.

The platform can check whether a notification has been processed by the OSPM either by polling the command complete bit, or where supported through receiving a doorbell interrupt from the OSPM. When the platform detects that the notification has been processed by the OSPM, the platform takes the following steps:

10. If polling check command complete is set. If using a doorbell this step is optional.
11. Processes the command response.

The platform must ensure that any writes in step 2 above are observable by the OSPM application processors before writes in step 3. Similarly, the OSPM must ensure that any writes in step 7 are observable by the platform before step 8 completes.

Individual protocols that use PCC define the meaning of notifications.

14.7 Referencing the PCC address space

An individual PCC register may be referenced by the Generic Address Structure or in a Generic Register Descriptor by using the Address Space ID PCC (0xA). When using the PCC address space, the Access Size field is redefined to Subspace ID, and identifies which PCC subspace the descriptor refers to.

As an example, the following resource template refers to the field occupying bits 8 through 15 at address 0x30 in PCC subspace 9:

```
ResourceTemplate()
{
    Register (
        PCC,                //AddressSpaceKeyword
        8,                  //RegisterBitWidth
        8,                  //RegisterBitOffset
        0x30,               //RegisterAddress
        9,                  //AccessSize (subspace ID)
    )
}
```

Note that the PCC address space may not be used in any resource template or register unless the register/resource field explicitly allows the use of the PCC address space.